

1 Lambda Equations

The first step to getting the EOM equations is the lambda equations for the ground state. This way, we don't need to solve for these along side the other parameters.

$$\frac{\partial L}{\partial t_i^a} = \langle \Phi_0 | [\tilde{H}_N, \hat{T}_2] + \tilde{H}_N | \Phi_i^a \rangle + \langle \Phi_0 | \hat{\Lambda} ([\tilde{F}_N, \hat{T}_2] + \tilde{H}_N) | \Phi_i^a \rangle - \lambda_i^a \langle \Phi_0 | [\tilde{F}_N, \hat{T}_2] + \tilde{H}_N | \Phi_0 \rangle = 0 \quad (1)$$

$$\frac{\partial L}{\partial t_{ij}^{ab}} = \langle \Phi_0 | [\tilde{F}_N, \hat{T}_2] + \tilde{H}_N | \Phi_{ij}^{ab} \rangle + \langle \Phi_0 | \hat{\Lambda}_1 [\bar{H}, \tau_{ij}^{ab}] | \Phi_0 \rangle + \langle \Phi_0 | \hat{\Lambda} \tilde{F}_N | \Phi_{ij}^{ab} \rangle = 0 \quad (2)$$

This gives the following residual equations. Note that we need to include disconnected diagrams between the lambda and Hamiltonian terms, since they are not commuted.

$$\langle \widetilde{ij|ab} \rangle + P_{ij} \sum_e \lambda_e^i \langle \widetilde{ej|ab} \rangle - P_{ab} \sum_m \lambda_a^m \langle \widetilde{ij|mb} \rangle + \varepsilon_{ij}^{ab} \lambda_{ab}^{ij} + P_{ij} P_{ab} \lambda_a^i \tilde{f}_{jb} = 0 \quad (3)$$

$$\begin{aligned} \tilde{f}_{ia} + \sum_e \lambda_e^i \tilde{f}_{ea} - \sum_m \lambda_a^m \tilde{f}_{im} + \sum_{me} \lambda_e^n \langle \widetilde{ie|am} \rangle + \frac{1}{2} \sum_{mef} \lambda_{ef}^{im} \langle \widetilde{ef|am} \rangle - \frac{1}{2} \sum_{mne} \lambda_{ae}^{mn} \langle \widetilde{ie|mn} \rangle + \sum_{mnef} \lambda_f^n t_{mn}^{ef} \langle \widetilde{im|ae} \rangle \\ - \frac{1}{2} \sum_{mnef} \lambda_e^i t_{mn}^{ef} \langle \widetilde{mn|af} \rangle - \frac{1}{2} \sum_{mnef} \lambda_a^m t_{mn}^{ef} \langle \widetilde{in|ef} \rangle = 0 \end{aligned} \quad (4)$$

1.1 Spin-integration

The spin-integrated equations are as follows.

$$2\tilde{L}_{ijab} + 2P_{ia,jb} \sum_e \lambda_e^i \tilde{L}_{ejab} - 2P_{ia,jb} \sum_m \lambda_a^m \tilde{L}_{ijmb} + 2\varepsilon_{ij}^{ab} (2\lambda_{ab}^{ij} - \lambda_{ba}^{ij}) + 2P_{ia,jb} (2\lambda_a^i \tilde{f}_{jb} - \tilde{\lambda}_a^j \tilde{f}_{ib}) = 0 \quad (5)$$

$$\begin{aligned} 2\tilde{f}_{ia} + 2 \sum_e \lambda_e^i \tilde{f}_{ea} - 2 \sum_m \lambda_a^m \tilde{f}_{im} + 2 \sum_{me} \lambda_e^m \tilde{L}_{ieam} + 2 \sum_{mef} \lambda_{ef}^{im} \langle \widetilde{ef|am} \rangle \\ - 2 \sum_{mne} \lambda_{ae}^{mn} \langle \widetilde{ie|mn} \rangle + 2 \sum_{mnef} \lambda_f^n (2t_{mn}^{ef} - t_{mn}^{fe}) \tilde{L}_{imae} - 2 \sum_{mnef} \lambda_e^i t_{mn}^{ef} \tilde{L}_{mnae} - 2 \sum_{mnef} \lambda_a^m t_{mn}^{ef} \tilde{L}_{inef} = 0 \end{aligned} \quad (6)$$

1.2 Alternative Spin-integration

We can alternatively split the biorthogonalization for the lambda equations. We will need to do this for EOM as well.

$$2\langle \widetilde{ij|ab} \rangle + 2P_{ia,jb} \sum_e \lambda_e^i \langle \widetilde{ej|ab} \rangle - 2P_{ia,jb} \sum_m \lambda_a^m \langle \widetilde{ij|mb} \rangle + 2\varepsilon_{ij}^{ab} \lambda_{ab}^{ij} + 2P_{ia,jb} \lambda_a^i \tilde{f}_{jb} = 0 \quad (7)$$

$$\begin{aligned} 2\tilde{f}_{ia} + 2 \sum_e \lambda_e^i \tilde{f}_{ea} - 2 \sum_m \lambda_a^m \tilde{f}_{im} + 2 \sum_{me} \lambda_e^m \tilde{L}_{ieam} + 2 \sum_{mef} (2\lambda_{ef}^{im} - \lambda_{fe}^{im}) \langle \widetilde{ef|am} \rangle \\ - 2 \sum_{mne} (2\lambda_{ae}^{mn} - \lambda_{ea}^{mn}) \langle \widetilde{ie|mn} \rangle + 2 \sum_{mnef} \lambda_f^n (2t_{mn}^{ef} - t_{mn}^{fe}) \tilde{L}_{imae} - 2 \sum_{mnef} \lambda_e^i t_{mn}^{ef} \tilde{L}_{mnae} - 2 \sum_{mnef} \lambda_a^m t_{mn}^{ef} \tilde{L}_{inef} = 0 \end{aligned} \quad (8)$$

1.3 Density-fitting

$$4 \sum_Q \tilde{b}_{ia}^Q \tilde{b}_{jb}^Q - 2 \sum_Q \tilde{b}_{ib}^Q \tilde{b}_{ja}^Q + 2 \varepsilon_{ij}^{ab} \lambda_{ab}^{ij} = 0 \quad (9)$$

$$\begin{aligned} 2\tilde{f}_{ia} + 2 \sum_e \lambda_e^i \tilde{f}_{ea} - 2 \sum_m \lambda_a^m \tilde{f}_{im} + \sum_{me} (2\lambda_{ae}^{im} - \lambda_{ea}^{im}) \tilde{f}_{em} + 4 \sum_Q \tilde{b}_{ia}^Q \sum_{me} \lambda_e^m \tilde{b}_{em}^Q - 2 \sum_{me} \lambda_e^m \sum_Q \tilde{b}_{im}^Q \tilde{b}_{ea}^Q \\ + 4 \sum_{Qe} \tilde{b}_{ea}^Q \sum_{mf} \lambda_{ef}^{im} \tilde{b}_{fm}^Q - 2 \sum_{fQ} \tilde{b}_{fa}^Q \sum_{me} \lambda_{ef}^{im} \tilde{b}_{em}^Q - 4 \sum_{mQ} \tilde{b}_{im}^Q \sum_{ne} \lambda_{ae}^{mn} \tilde{b}_{en}^Q + 2 \sum_{nQ} \tilde{b}_{in}^Q \sum_{me} \lambda_{ae}^{mn} \tilde{b}_{em}^Q = 0 \end{aligned} \quad (10)$$

2 One-particle reduced density matrix

The one-particle reduced density matrix in the T_1 transformation formalism is as follows.

$$\tilde{\rho}_{ij} = \delta_{ij} - \frac{1}{2} \sum_{mef} \lambda_{ef}^{jm} t_{im}^{ef} \quad (11)$$

$$\tilde{\rho}_{ab} = \frac{1}{2} \sum_{mne} \lambda_{ae}^{mn} t_{mn}^{be} \quad (12)$$

$$\tilde{\rho}_{ai} = \sum_{me} \lambda_e^m t_{im}^{ae} \quad (13)$$

$$\tilde{\rho}_{ia} = \lambda_a^i \quad (14)$$

The T_1 transformed property matrix has the following form.

$$\tilde{\theta}_{ij} = (1 - \delta_{ij}) \theta_{ij} + \sum_e t_j^e \theta_{ie} \quad (15)$$

$$\tilde{\theta}_{ab} = (1 - \delta_{ab}) \theta_{ab} - \sum_m t_m^a \theta_{mb} \quad (16)$$

$$\tilde{\theta}_{ia} = \theta_{ia} \quad (17)$$

$$\tilde{\theta}_{ai} = \theta_{ai} + \sum_e t_i^e \tilde{\theta}_{ae} - \sum_m t_m^a \theta_{mi} \quad (18)$$

The property can then be found as $\theta = \text{Tr}(\tilde{\rho}\tilde{\theta}) = \sum_{pq} \tilde{\rho}_{pq} \tilde{\theta}_{pq}$. Alternatively, we can untransform the equations to get the following. Note that the signs are not as expected, due to the fact that we need to do an inverse T_1 transform.

$$\rho_{ij} = \tilde{\rho}_{ij} - \delta_{ij} - \sum_e t_i^e \tilde{\rho}_{je} \quad (19)$$

$$\rho_{ab} = \tilde{\rho}_{ab} + \sum_m t_m^b \tilde{\rho}_{ma} \quad (20)$$

$$\rho_{ai} = \tilde{\rho}_{ai} - \sum_e t_i^e \rho_{ea} + \sum_m t_m^a \tilde{\rho}_{im} \quad (21)$$

$$\rho_{ia} = \tilde{\rho}_{ia} \quad (22)$$

2.1 Spin integration

$$\rho_{ij} = -2 \sum_e \lambda_e^j t_i^e - 2 \sum_{mef} (2\lambda_{ef}^{jm} - \lambda_{fe}^{jm}) t_{im}^{ef} \quad (23)$$

$$\rho_{ab} = 2 \sum_m \lambda_a^m t_m^b + 2 \sum_{mne} (2\lambda_{ae}^{mn} - \lambda_{ea}^{mn}) t_{mn}^{be} \quad (24)$$

$$\rho_{ai} = 2t_i^a + 2 \sum_{me} \lambda_e^m (2t_{im}^{ae} - t_{im}^{ea}) - 2 \sum_{me} \lambda_e^m t_i^e t_m^a - 2 \sum_{mnef} \lambda_{ef}^{mn} t_i^e t_{mn}^{af} - 2 \sum_{mnef} \lambda_{ef}^{mn} t_m^a t_{in}^{ef} \quad (25)$$

$$\rho_{ia} = 2\lambda_a^i \quad (26)$$

2.2 Alternative Spin-integration

$$\tilde{\rho}_{ij} = 2\delta_{ij} - 2 \sum_{mef} (2\lambda_{ef}^{jm} - \lambda_{fe}^{jm}) t_{im}^{ef} \quad (27)$$

$$\tilde{\rho}_{ab} = 2 \sum_{mne} (2\lambda_{ae}^{mn} - \lambda_{ea}^{mn}) t_{mn}^{be} \quad (28)$$

$$\tilde{\rho}_{ia} = 2\lambda_a^i \quad (29)$$

$$\tilde{\rho}_{ai} = 2 \sum_{me} \lambda_e^m (2t_{im}^{ae} - t_{im}^{ea}) \quad (30)$$

$$\rho_{ij} = -2 \sum_{mef} (2\lambda_{ef}^{jm} - \lambda_{fe}^{jm}) t_{im}^{ef} - 2 \sum_e t_i^e \lambda_e^j \quad (31)$$

$$\rho_{ab} = 2 \sum_{mne} (2\lambda_{ae}^{mn} - \lambda_{ea}^{mn}) t_{mn}^{be} + 2 \sum_m t_m^b \lambda_a^m \quad (32)$$

$$\rho_{ai} = 2 \sum_{me} \lambda_e^m (2t_{im}^{ae} - t_{im}^{ea}) - 2 \sum_{mnef} (2\lambda_{ef}^{mn} - \lambda_{fe}^{mn}) t_i^e t_{mn}^{af} - 2 \sum_{me} \lambda_e^m t_i^e t_m^a + 2t_i^a - 2 \sum_{mnef} (2\lambda_{ef}^{mn} - \lambda_{fe}^{mn}) t_m^a t_{in}^{ef} \quad (33)$$

$$\rho_{ia} = 2\lambda_a^i \quad (34)$$

3 Right Products

For the ground state, $r_0 = 1$ and $r_\mu = 0$. For excited states, all amplitudes need to be solved.

$$\sum_{ia} r_i^a \tilde{f}_{ia} + \frac{1}{4} \sum_{ijab} r_{ij}^{ab} \langle ij || ab \rangle = \omega r_0 \quad (35)$$

$$\begin{aligned} \omega r_i^a = & \sum_e r_i^e \tilde{f}_{ae} - \sum_m r_m^a \tilde{f}_{mi} - \frac{1}{2} \sum_{mnef} r_i^e t_{mn}^{af} \langle mn || ef \rangle - \frac{1}{2} \sum_{mnef} r_m^a t_{in}^{ef} \langle mn || ef \rangle + \sum_{mnef} r_n^f t_{im}^{ae} \langle mn || ef \rangle \\ & + \sum_{me} r_{im}^{ae} \tilde{f}_{me} + \frac{1}{2} \sum_{mef} r_{im}^{ef} \langle am || ef \rangle - \frac{1}{2} \sum_{mne} r_{mn}^{ae} \langle mn || ie \rangle + \sum_{me} r_m^e \langle am || ie \rangle \end{aligned} \quad (36)$$

$$\omega r_{ij}^{ab} = \varepsilon_{ij}^{ab} r_{ij}^{ab} + P_{ij} \sum_e r_i^e \langle ab || ej \rangle - P_{ab} \sum_m r_m^a \langle mb || ij \rangle \quad (37)$$

3.1 Spin-integration

$$2 \sum_{ia} r_i^a \tilde{f}_{ia} + \sum_{ijab} r_{ij}^{ab} \tilde{L}_{ijab} = \omega r_0 \quad (38)$$

$$\begin{aligned} 2\omega r_i^a = & 2 \sum_e r_i^e \tilde{f}_{ea} - 2 \sum_m r_m^a \tilde{f}_{mi} + 2 \sum_{me} r_m^e \tilde{L}_{amie} + 2 \sum_{mnef} r_n^f (2t_{im}^{ae} - t_{im}^{ea}) \tilde{L}_{mnef} + 2 \sum_{me} r_{im}^{ae} \tilde{f}_{me} \\ & - 2 \sum_{mnef} r_i^e t_{mn}^{af} \tilde{L}_{mnef} - 2 \sum_{mnef} r_m^a t_{in}^{ef} \tilde{L}_{mnef} + 2 \sum_{mef} r_{im}^{ef} \tilde{L}_{amef} - 2 \sum_{mne} r_{mn}^{ae} \tilde{L}_{mnie} \end{aligned} \quad (39)$$

$$2\omega r_{ij}^{ab} = 2\varepsilon_{ij}^{ab} r_{ij}^{ab} + 2P_{ia,jb} \sum_e r_i^e \langle \widetilde{ab|ej} \rangle - 2P_{ia,jb} \sum_m r_m^a \langle \widetilde{mb|ij} \rangle \quad (40)$$

Let's find the singles-singles block as well.

$$H_{ia,jb} = 2\delta_{ij} \tilde{f}_{ba} - 2\delta_{ab} \tilde{f}_{ji} + 2\tilde{L}_{ajib} + 2 \sum_{me} (2t_{im}^{ae} - t_{im}^{ea}) \tilde{L}_{mjeb} - 2\delta_{ij} \sum_{mne} t_{mn}^{ae} \tilde{L}_{mnbe} - 2\delta_{ab} \sum_{mef} t_{im}^{ef} \tilde{L}_{jme f} \quad (41)$$

And the diagonal elements.

$$H_{ia,ia} = 2\tilde{f}_{aa} - 2\tilde{f}_{ii} + 2\tilde{L}_{aiaa} + 2 \sum_{me} (2t_{im}^{ae} - t_{im}^{ea}) \tilde{L}_{miea} - 2 \sum_{mne} t_{mn}^{ae} \tilde{L}_{mnae} - 2 \sum_{mef} t_{im}^{ef} \tilde{L}_{ime f} \quad (42)$$

$$H_{ijab,ijab} = 2\varepsilon_{ij}^{ab} \quad (43)$$

4 Left Products

Similar to the right products, for the ground state, $l_0 = 1$, and overall $\hat{L} = 1 + \hat{\Lambda}$. For excited states, $l_0 = 0$. Also, unlike for the right-hand case, the left-hand case is allowed to have disconnected diagrams because the left-hand multiplication is not done with a commutator.

$$\begin{aligned} \omega l_a^i = & \sum_e l_e^i \tilde{f}_{ea} - \sum_m l_a^m \tilde{f}_{im} + \sum_{me} l_e^m \langle \widetilde{ie||am} \rangle - \frac{1}{2} \sum_{mnef} l_a^m t_{mn}^{ef} \langle \widetilde{in||ef} \rangle - \frac{1}{2} \sum_{mnef} l_e^i t_{mn}^{ef} \langle \widetilde{mn||af} \rangle \\ & + \frac{1}{2} \sum_{mef} l_{ef}^{im} \langle \widetilde{ef||am} \rangle - \frac{1}{2} \sum_{mne} l_{ae}^{mn} \langle \widetilde{ie||mn} \rangle + \sum_{mnef} l_f^m t_{mn}^{ef} \langle \widetilde{im||ae} \rangle \end{aligned} \quad (44)$$

$$\omega l_{ab}^{ij} = P_{ij} \sum_e l_e^i \langle \widetilde{ej||ab} \rangle - P_{ab} \sum_m l_a^m \langle \widetilde{ij||mb} \rangle + \varepsilon_{ij}^{ab} l_{ab}^{ij} + P_{ij} P_{ab} l_a^i \tilde{f}_{jb} \quad (45)$$

4.1 Spin-integrated

$$\begin{aligned} 2\omega l_a^i = & 2 \sum_e l_e^i \tilde{f}_{ea} - 2 \sum_m l_a^m \tilde{f}_{im} + 2 \sum_{me} l_i^m \tilde{L}_{ieam} - 2 \sum_{mnef} l_a^m t_{mn}^{ef} \tilde{L}_{inef} - 2 \sum_{mnef} l_e^i t_{mn}^{ef} \tilde{L}_{mna f} \\ & + 2 \sum_{mef} l_{ef}^{im} \langle \widetilde{ef||am} \rangle - 2 \sum_{mne} l_{ae}^{mn} \langle \widetilde{ie||mn} \rangle + 2 \sum_{mnef} l_f^m (2t_{mn}^{ef} - t_{mn}^{fe}) \tilde{L}_{ima e} \end{aligned} \quad (46)$$

$$2\omega \left(2l_{ab}^{ij} - l_{ba}^{ij} \right) = 2P_{ia,jb} \sum_e l_e^i \tilde{L}_{ejab} - 2P_{ia,jb} \sum_m l_a^m \tilde{L}_{ijmb} + 2\varepsilon_{ij}^{ab} \left(2l_{ab}^{ij} - l_{ba}^{ij} \right) + 2P_{ia,jb} \left(2l_a^i \tilde{f}_{jb} - l_b^i \tilde{f}_{ja} \right) \quad (47)$$

That won't work, so we need to fix it.

$$\frac{2}{3}\omega \left(4l_{ab}^{ij} - 2l_{ba}^{ij} + 2l_{ba}^{ij} - l_{ab}^{ij} \right) = 2\omega l_{ab}^{ij} \quad (48)$$

$$6\omega l_{ab}^{ij} = 2P_{ia,jb} \sum_e l_e^i \left(2\tilde{L}_{ejab} + \tilde{L}_{ejba} \right) - 2P_{ia,jb} \sum_m l_a^m \left(2\tilde{L}_{ijmb} + \tilde{L}_{jimb} \right) + 2\varepsilon_{ij}^{ba} \left(4l_{ab}^{ij} - 2l_{ba}^{ij} + 2l_{ba}^{ij} - l_{ab}^{ij} \right) \\ + 2P_{ia,jb} \left(4l_a^i \tilde{f}_{jb} - 2l_b^i \tilde{f}_{ja} + 2l_b^i \tilde{f}_{ja} - l_a^i \tilde{f}_{jb} \right) \quad (49)$$

$$2\omega l_{ab}^{ij} = 2P_{ia,jb} \sum_e l_e^i \langle e j | a b \rangle - 2P_{ia,jb} \sum_m l_a^m \langle i j | m b \rangle + 2\varepsilon_{ij}^{ab} l_{ab}^{ij} + 2P_{ia,jb} l_a^i \tilde{f}_{jb} \quad (50)$$

$$2\omega l_a^i = 2 \sum_e l_e^i \tilde{f}_{ea} - 2 \sum_m l_a^m \tilde{f}_{im} + 2 \sum_{me} l_e^m \tilde{L}_{ieam} - 2 \sum_{mnef} l_a^m t_{mn}^{ef} \tilde{L}_{inef} - 2 \sum_{mnef} l_e^i t_{mn}^{ef} \tilde{L}_{mnae} \\ + 2 \sum_{mef} (2l_{ef}^{im} - l_{fe}^{im}) \langle e f | a m \rangle - 2 \sum_{mne} (2l_{ae}^{mn} - l_{ea}^{mn}) \langle i e | m n \rangle + 2 \sum_{mnef} l_f^n (2t_{mn}^{ef} - t_{mn}^{fe}) \tilde{L}_{imae} \quad (51)$$

5 Dot products

The dot product between two vectors is as follows. Note that these expressions are independent on whether the two vectors are from the same state, meaning these can be used to orthonormalize the basis of states.

$$\langle \Phi_0 | \hat{L} \hat{R} | \Phi_0 \rangle = \sum_{ia} l_a^i r_i^a + \frac{1}{4} \sum_{ijab} l_{ab}^{ij} r_{ij}^{ab} \quad (52)$$

$$\langle \Phi_0 | (1 + \hat{\Lambda}) \hat{R} | \Phi_0 \rangle = r_0 + \sum_{ia} \lambda_a^i r_i^a + \frac{1}{4} \sum_{ijab} \lambda_{ab}^{ij} r_{ij}^{ab} \quad (53)$$

$$\langle \Phi_0 | \hat{R}^\dagger \hat{R} | \Phi_0 \rangle = |r_0|^2 + \sum_{ia} |r_i^a|^2 + \frac{1}{4} \sum_{ijab} |r_{ij}^{ab}|^2 \quad (54)$$

$$\langle \Phi_0 | \hat{L} \hat{L}^\dagger | \Phi_0 \rangle = \sum_{ia} |l_a^i|^2 + \frac{1}{4} \sum_{ijab} |l_{ab}^{ij}|^2 \quad (55)$$

5.1 Spin-integration

The transformation matrix from the manifold of left-excitations to the spin-adapted left-excitations is baked into the $\hat{L}$ operator, so it only needs to be considered for the same-vector dot products. It will be denoted $\hat{U}$.

$$\langle \Phi_0 | \hat{L} \hat{R} | \Phi_0 \rangle = 2 \sum_{ia} l_a^i r_i^a + 2 \sum_{ijab} l_{ab}^{ij} r_{ij}^{ab} \quad (56)$$

$$\langle \Phi_0 | (1 + \hat{\Lambda}) \hat{R} | \Phi_0 \rangle = r_0 + 2 \sum_{ia} \lambda_a^i r_i^a + 2 \sum_{ijab} \lambda_{ab}^{ij} r_{ij}^{ab} \quad (57)$$

$$\langle \Phi_0 | \hat{R}^\dagger \hat{U} \hat{R} | \Phi_0 \rangle = |r_0|^2 + 2 \sum_{ia} |r_i^a|^2 + 2 \sum_{ijab} |r_{ij}^{ab}|^2 \quad (58)$$

$$\langle \Phi_0 | \hat{L} \hat{U}^{-1} \hat{L}^\dagger | \Phi_0 \rangle = 2 \sum_{ia} |l_a^i|^2 + 2 \sum_{ijab} |l_{ab}^{ij}|^2 \quad (59)$$